

Spectral Delegation for Anonymous DHT Lookups

From mixing certificates to receipt-grade per-step budgets

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Abstract

Random-walk delegation is a standard building block for “anonymous” DHT lookups: a querier hands work to a relay chosen via a short walk on an overlay, and the relay performs the visible step. The published spectral anonymity note gives sharp distinguishing bounds for delegation in (possibly directed) overlays. This paper extracts a *receipt-grade* corollary: a compact spectral witness (e.g. a second-singular-value bound) can be turned into an auditable per-step MaxL (bits) cap; via the endpoint bridge this immediately yields an odds-inflation bound and identification sizing rules[2] that fits directly into the Regime II conditional-certificate interface. Mechanism papers should cite this note (and the published spectral note) rather than re-deriving spectral-to-budget translations.

Scope and non-redundancy. We only do the “spectral $\Rightarrow$ per-step budget” translation. Receipt schema, replay hooks, and adaptive horizon accounting are handled elsewhere.[5, 6, 7] Systems and related-work context for anonymous DHT delegation is indexed in Synthesis 7.[3]

Canonical backbone. The full spectral anonymity result and optimal distinguishing bounds are owned by the published mathematics note: [[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]].

Units. Budgets are measured in bits: we use $\log_2$ and 2^ϵ for odds-inflation / pointwise-maximal-leakage style caps.[1]

1 Delegation model and spectral witness

We adopt the delegation setting of the published spectral anonymity paper.[4] A start node S (the querier, or the querier’s entry relay) initiates a length- t delegation walk on an n -node overlay. The observed delegate D is the terminal node of the walk (or, more generally, the node that performs a visible lookup step).

The published note defines a discriminant operator for possibly directed kernels and shows that distinguishing power contracts at rate controlled by a second singular value $\sigma_2 \in [0, 1)$ of that operator (or a conservative certificate upper-bounding it).[4] For this paper, the *witness* is simply: the overlay size n , the walk length t , the kernel identifier (a digest), and an auditor-checkable bound σ_2 .

Remark 1 (Why a spectral witness fits receipts). *A receipt cannot enumerate a huge conditioning space (all possible route histories), but it can commit to a small, digest-bound kernel and a small*

numerical witness. Regime II conditional certificates (Synthesis 21) are designed to carry exactly that shape: a conditioning interface identifier plus a compact evidence object.[6]

2 A conservative spectral-to-budget translation

The published spectral note provides (among stronger results) a conservative bound on expected χ^2 leakage of the posterior on S given D . [4] We record a short corollary that yields an auditable MaxL cap in bits with a tail parameter δ .

Proposition 1 (Spectral contraction $\Rightarrow$ a high-probability per-step MaxL bound (uniform prior)). *Assume the stationary prior π is uniform on n nodes. Let σ_2 be an upper bound on the second singular value of the discriminant operator for the delegation kernel (as defined in the published spectral note).[4] Let $\mu_D(\cdot) \stackrel{\text{def}}{=} \mathbb{P}[S = \cdot \mid D]$ be the posterior on S given the observed delegate D . Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$ over $D \sim \pi$,*

$$\mathcal{L}(S \rightarrow D) \leq \log_2 \left(1 + \sqrt{n(n-1)} \sigma_2^t / \sqrt{\delta} \right) \text{ bits.}$$

Proof sketch. The published spectral note gives $\mathbb{E}_{D \sim \pi} [\chi^2(\mu_D \parallel \pi)] \leq (n-1) \sigma_2^{2t} \cdot [4]$ Markov's inequality yields $\chi^2(\mu_D \parallel \pi) \leq (n-1) \sigma_2^{2t} / \delta$ with probability at least $1 - \delta$. For uniform π , $\chi^2(\mu \parallel \pi) = n \sum_s (\mu(s) - 1/n)^2$, so $\|\mu - \pi\|_2 = \sqrt{\chi^2(\mu \parallel \pi) / n}$ and $\|\mu - \pi\|_\infty \leq \|\mu - \pi\|_2$. Thus $\max_s \mu(s) \leq 1/n + \sqrt{\chi^2(\mu \parallel \pi) / n}$, implying $\max_s \mu(s) / \pi(s) \leq 1 + \sqrt{n \chi^2(\mu \parallel \pi)}$. Taking $\log_2$ gives the stated bound. $\square$

Remark 2 (What this bound is (and is not)). *This corollary is intentionally conservative and assumes a uniform stationary prior. It is nevertheless useful as a receipt-side auditable cap: the parameters (n, t, σ_2, δ) are small, and the auditor can recompute the right-hand side. Sharper and more general statements (non-uniform priors, directed kernels, optimal tests) live in the published spectral note.[4]*

Proposition 2 (From a χ^2 witness to a MaxL cap (non-uniform priors)). *Let π be any prior on S with $\pi_{\min} \stackrel{\text{def}}{=} \min_s \pi(s) > 0$. For any posterior μ on S ,*

$$\max_s \frac{\mu(s)}{\pi(s)} \leq 1 + \sqrt{\chi^2(\mu \parallel \pi) / \pi_{\min}}.$$

Consequently, if an evidence object implies that with probability at least $1 - \delta$ (over D) we have $\chi^2(\mu_D \parallel \pi) \leq B$, then with the same probability

$$\mathcal{L}(S \rightarrow D) \leq \log_2 \left(1 + \sqrt{B / \pi_{\min}} \right) \text{ bits.}$$

Proof sketch. By definition, $\chi^2(\mu \parallel \pi) = \sum_s (\mu(s) - \pi(s))^2 / \pi(s)$, so for each s , $(\mu(s) - \pi(s))^2 / \pi(s) \leq \chi^2(\mu \parallel \pi)$ and hence $|\mu(s) - \pi(s)| \leq \sqrt{\pi(s) \chi^2(\mu \parallel \pi)}$. Therefore $\mu(s) / \pi(s) \leq 1 + \sqrt{\chi^2(\mu \parallel \pi) / \pi(s)} \leq 1 + \sqrt{\chi^2(\mu \parallel \pi) / \pi_{\min}}$. Take $\log_2$. $\square$

3 Receipt-side use: Regime II conditional certificates

A deployment that uses delegation to hide a querier's identity (or entry relay) can publish a Regime II per-step conditional certificate whose evidence object is:

`{ kernel_digest, n, t, sigma2_upper, delta }.`

The replay plan commits to the kernel digest and the numerical witness. An auditor recomputes the per-step cap b_t from proposition 1 and records it as the checked conditional budget for that step. Adaptive horizon accounting then composes the checked per-step caps additively (Synthesis 19), without assuming independence.[7, 6]

Where to put the witness. This paper does not add new receipt fields. The evidence object lives behind the existing replay hook (plan identifier + plan-spec digest), so the public receipt remains short.[8]

Non-uniform priors (class budgets). If the operator budgets “user within a class” (non-uniform π), keep the receipt surface unchanged: include a prior certificate parameter such as $\pi_{\min}$ and use proposition 2 to translate any χ^2 witness into a MaxL cap. The published spectral note contains the sharper (and directed-kernel) statements that supply such witnesses; this paper only packages the final receipt-side conversion.[4]

4 Research warranted

The most useful refinements are interface refinements, not longer papers: (i) non-uniform priors and class-based priors (so the budget targets “user within a set”); and (ii) directed overlays and time-varying kernels (so the kernel digest is a schedule rather than a single matrix). Both are treated in the published spectral note at the level of optimal distinguishing bounds;[4] the remaining work is to package those bounds into compact, auditor-friendly certificates consistent with Regime II.

References

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- [4] Anonymous. Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays. Published note, [[2026.01.23 - Mathematics: Spectral Anonymity and Optimal Distinguishing Bounds for Random-Walk Delegation in (Possibly Directed) Overlays]], 2026. (This archive.)
- [5] Anonymous. Receipt Line Items for Anonymity Claims: A Minimal Tuple Schema for Published Budgets. Synthesis 12, The-One-True-Archive (2026).
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